Notes_25.03: Understanding the Confusion Matrix







🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

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📌 Executive Summary

2x2 Contingency table for binary classification, Type I and Type II errors.

🎯 Mathematical Formulations

Confusion Matrix Table:

$$\begin{array}{c|cc} & \text{Predicted Negative (0)} & \text{Predicted Positive (1)} \\ \hline \text{Actual Negative (0)} & \text{True Negative (TN)} & \text{False Positive (FP) [Type I Error]} \\ \text{Actual Positive (1)} & \text{False Negative (FN) [Type II Error]} & \text{True Positive (TP)} \end{array}$$

💡 Conceptual Principles & Architecture

4 Outcomes Ka Desi Matlab:

1. **True Positive (TP):** Mareez ko bimaari thi, aur model ne bhi bola bimaari hai (Sahi pakda!).
2. **True Negative (TN):** Insaan bilkul healthy tha, aur model ne bola healthy hai (Sahi chhora!).

3. **False Positive (FP - Type I Error):** Insaan healthy tha, lekin model ne jhootha alarm baja diya ki bimaari hai.
4. **False Negative (FN - Type II Error):** Mareez ko bimaari thi, lekin model ne bol diya ki healthy hai. (Medical field mein yeh sabse zyada deadly aur khatarnak error hota hai!).

💻 Python Production Implementation

```
from sklearn.metrics import confusion_matrix
import numpy as np

y_true = np.array([1, 0, 1, 1, 0, 1, 0, 0, 1, 0])
y_pred = np.array([1, 0, 1, 0, 0, 1, 1, 0, 1, 0])

cm = confusion_matrix(y_true, y_pred)
tn, fp, fn, tp = cm.ravel()

print("Confusion Matrix:\n", cm)
print(f"TP: {tp}, TN: {tn}, FP: {fp}, FN: {fn}")
```